Finite-length and vertical effects in nonlinear-kicker injection into a Korea Light Source design lattice

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Abstract

We quantify the error of a single thin kick map when the beam is strongly deflected inside a nonlinear injection kicker, using established segmented tracking through the conductor field. For a 0.525 m four-loop magnet in a 4 GeV Korea Light Source design lattice, we compare tracking in an exact filament field, thin and sliced vector maps, and an independent adaptive trajectory solver. In a diagnostic launch family, the thin-map error grows empirically as $\eta^{1.46}$ with the ratio η of the trajectory sagitta to the kick-lobe width. Along the matched trajectory of the selected operating point, the error is 161 μrad, 3.7 times a tenth of the angular window. A sliced vector map agrees with the exact field. With the design booster beam, a declared chamber and an operating point selected on ten corrected rings, capture is 98.4 % at 500 turns on the ideal ring and 78.0 % on average over the corrected rings. Under joint machine and shot errors the mean falls to 16.1 %, with 51 % of the realizations capturing nothing, against 44.1 % for an equally strong uniform kick. A paired error-class ablation traces the deficit without booster jitter to transfer-line magnet errors. These leave a 0.90 mm rms vertical injection error that the model's horizontal-only steering does not remove. Removing this error in a declared sensitivity raises the NKM to 76.6 %, after which booster jitter becomes the NKM-specific loss, with comparable position and angle contributions. The deficit also depends on the operating point: at two alternative points NKM capture is unchanged while the uniform kick loses 17–23 points. Ideal centroid cancellation does not make the magnet transparent; for 10 μm rms wire placement the stored beam is displaced by several rms sizes, so compensation is required. These are conditional simulation results.

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I. INTRODUCTION

Fourth-generation storage rings reduce the electron emittance by an order of magnitude or more with multibend-achromat lattices [1, 2]. The required strong focusing and sextupoles leave a small dynamic aperture (DA), the main constraint on injection. These rings also run in top-up mode [3]: the beam is replenished every few minutes during user operation, so each injection should leave the stored beam undisturbed. Where the DA is too small for off-axis injection, the beam can be injected on axis by swapping bunches with an accumulator [4] or injected into the longitudinal acceptance [5]. Where off-axis injection remains possible, the conventional four-kicker orbit bump leaves residual oscillations because the kicker pulses cannot be matched perfectly.

A single pulsed multipole avoids residual oscillations from mismatched kicker pulses by making its field vanish on the stored-beam axis. Injection with a pulsed quadrupole was demonstrated at the KEK PF-AR [6] and with a pulsed sextupole at the KEK Photon Factory [7]. A non-linear kicker (NLK) with an off-axis field maximum was developed at BESSY II [8, 9], and a multipole injection kicker (MIK) met a transparency specification of 10 % of the rms beam size at MAX IV and reached 2 % at SOLEIL [10–12]. At MAX IV, the difficulty of commissioning pulsed-multipole injection led to the addition of a single dipole kicker for commissioning [13]. At SIRIUS, NLK injection was predicted to reach 99 % efficiency for a -9 mm DA but delivered about 85 ± 8 % with the design sextupoles, and online sextupole optimisation was needed [14, 15]; BESSY II likewise reached up to 97 % only with optimised sextupoles [9].

The kick varies across the injected bunch even when it vanishes on the stored-beam axis. Liu *et al.* [14] identified this nonlinearity and the injected-beam size as the main efficiency limit, which motivates placing the bunch near the kick extremum. Design studies usually describe the kicker by a kick map, the straight-line field integral applied at the magnet centre [16]. A single thin map requires a finite-length accuracy check: the injected beam is deflected by several milliradians inside the magnet, and the vertical kick off the midplane follows from the field's curl-free structure, which requires a consistently signed vector map. Integrating the field along the actual path, as is done for insertion devices and fringe fields [17], resolves trajectory displacement while preserving the physical vector field. Segmented field tracking has also been used for nonlinear kickers: Sun *et al.* [18]

optimized an eight-conductor ALS kicker with segmented drift–kick–drift tracking through a Biot–Savart field, checked against Runge–Kutta integration and a finite three-dimensional conductor model with return loops, and found that the finite-conductor fringe field did not significantly affect injection efficiency. That kicker is 0.28 m long, receives the beam almost parallel (0.08 mrad at the septum) and kicks by about 1 mrad, as estimated from the published field profile; its trajectory sagitta inside the magnet is therefore of order 0.04 mm.

The Korea Light Source (KLS) is a 4 GeV, 28-cell hybrid multibend-achromat ring of about 800 m circumference with a natural emittance of 62 pm [19]. This article studies these effects for a 0.525 m four-loop wire NKM in a KLS design lattice, with the chamber-limited acceptance reported below. The kicker is modelled by the exact three-dimensional field of its current filaments, validated against the RADIA [20] kick map, and tracked through in slices. Using these established tracking tools, we quantify the approximation error in a regime with stronger self-deflection: the kicker cancels a 4.85 mrad incoming angle within 0.525 m, so the sagitta (0.32 mm) is a few percent of the kick-lobe width. In this regime a single thin map shifts the matched angle and changes the sign-dependent error response, while a sliced vector map does not. The sagitta-to-lobe-width ratio organizes the thin-map error over two wire families; the ALS sagitta is about an order of magnitude smaller than here, consistent with its published finding. We also check off-midplane trajectories independently, evaluate conditional capture and transparency tolerances, and use paired comparisons to attribute capture loss under errors. Vector maps require a consistently signed vertical field; this requirement applies equally to established kick-map methods. The study includes a declared chamber and ten corrected-error rings; it does not predict commissioned-machine performance.

The BESSY hardware values describe the 2011 design; the quoted efficiency was reported in 2023. These values are from different dates. SOLEIL's active compensation and measured construction errors demonstrate why geometric symmetry alone is insufficient [12]. The present tolerance study addresses idealized single-error sensitivities rather than reproducing those machines.

TABLE I. Selected published multipole-injection systems and this simulation. Top: hardware dimensions and a quoted kick (not necessarily a peak). Bottom: acceptance, efficiency and normalized disturbance under their original conditions; a dash means no comparable value in the cited source. Different lattices and efficiency definitions prevent a ranking.

System / type	Magnetic length [m]	Kick [mrad]	Physical aperture [mm]
PF / sextupole [7]	0.300	1.44 at $x = 15$ mm	66 bore diameter
BESSY II / NLK [8]	0.280	1.0 at $x = 12$ mm	42×10
ALS / NLK [18]	0.280	—	32×11
MAX IV / MIK [10]	—	—	—
SOLEIL / MIK [12]	0.304	2.2 nominal	46.8×7.8
SIRIUS / NLK [14, 15]	—	—	—
This study / four-loop NKM	0.525	4.89 peak (earlier $S = 0.85$)	32×7.6 assumed

System	Horizontal DA	Capture / efficiency	Stored-beam disturbance
PF	—	—	—
BESSY II	—	up to 97% with optimized sextupoles [9]	—
ALS	—	about 90% simulated [18]	—
MAX IV	—	—	below 10% rms criterion [12]
SOLEIL	$-13/+11$ mm at injection	97% peak, bare lattice	below 2% rms with active compensation [12]
SIRIUS	design -9 mm	99% simulated (213 turns); about 85% measured, 8% variability	—
This study	first lost grid: - 11.0/12.5 mm	98.4% simulated on the ideal ring (500 turns); 16.1% mean under the assumed joint errors	0.12 of σ_x at 0.2 mm horizontal offset, ideal geometry, $S = 0.95$

II. SYSTEM AND MODEL

A. Storage ring

The supplied file `K4GSR_HBIv4-1` is identified by the author as the lattice used by Kim *et al.* at IPAC'25 [19]. It is a design version, not the latest KLS lattice; its file identifier is retained because an independent release date was not supplied. Later lattice versions may change the nonlinear acceptance, so results here apply to this version.

The KLS design lattice (file `K4GSR_HBIv4-1`; source circumference 799.30 m, 3481 elements) is tracked with the Accelerator Toolbox [21, 22] in six dimensions with RF and deterministic radiation damping (table II). No stochastic quantum-excitation pass is included. The ring is re-ordered to start at the NKM centre, which is the kick and observation point. The synchrotron period is about 290 turns and the horizontal damping time about 3950 turns. The final capture studies include the declared chamber and septum blade as well as dynamical losses (section V). Figure 1 isolates the supplied lattice and septum without the additional chamber profile: on momentum, particles with $x' = 0$ are stable between -12 and $+13$ mm. Throughout, the *first unstable launch* on a side is the smallest $|x|$ on the launch grid (1 mm here, 0.5 mm for the corrected rings) whose particle is lost within 500 turns; it locates the stable edge to within one grid step and does not exclude stable islands beyond it.

B. Kicker field and tracking

The 0.525 m four-loop magnet is a preliminary design for this study by Dong-Eon Kim (Pohang Accelerator Laboratory), as stated by the author; the KLS baseline uses a different injection kicker. The baseline uses a four-kicker orbit bump and a 1 mm thin septum [23]. The alternative NKM scheme studied here retains an assumed 2 mm blade and a stored-beam edge at -16 mm; these dimensions define the study geometry and differ from the published baseline septum. Loop powering has not been specified, so series and independent-loop cases are distinguished in the transparency tolerances.

The NKM consists of four closed rectangular filament loops, one per quadrant, each with an inner and an outer longitudinal wire joined by end connectors (figure 2, left). The field of straight filaments in vacuum is evaluated in closed form (Biot–Savart), including

TABLE II. Principal parameters of the supplied design lattice and study geometry. Booster emittance and energy spread are design values; extraction Twiss, bunch length and vertical ratio are assumptions.

Ring energy, circumference	4 GeV, 799.30 m
Tunes, chromaticities	68.18/23.26, +6.0/ + 3.4
Emittance, energy spread	61.6 pm rad, 1.26×10^{-3}
Optics at NKM centre	$\beta_x = 16.27$ m, $\alpha_x = -0.145$, $\beta_y = 5.60$ m, $D_x = 0.3$ mm
Stored-beam size at NKM	$\sigma_x = 31.7\,\mu$m; $\varepsilon_y = 0.1\,\varepsilon_x$ assumed
Booster extraction Twiss	$\beta_{x,y} = 7.56/12.27$ m
Booster beam at extraction	$\varepsilon_{x,y} = 7.67/0.767$ nm rad, $\sigma_\delta = 1.06 \times 10^{-3}$
Injected beam at septum exit	$D_x = 80.9$ mm, $D'_x = 47.5$ mrad from the BTS
RF frequency, total voltage	499.593 MHz, 3.08 MV
Momentum compaction, synchrotron tune	7.77×10^{-5}, 3.44×10^{-3}
Damping times ($x/y/\delta$)	10.54/19.43/16.79 ms
Rms bunch length (stored / injected)	3.63/13.4 mm
Reference arc / centre-bend half-apertures	12×9 / 12×4.87 mm
Assumed half-apertures: NKM chamber / injection straight / open IVU	16×3.8 / 30×15 / 12×9 mm
Pulse example	5.3 μs half-sine, injection at peak (analytic)
Transfer line (BTS)	21.79 m, matched optics (peak $\beta_{x,y} = 31.4/50.7$ m)
Septum	stored-beam edge −16 mm, blade 2 mm, 2.26 m upstream of NKM centre
NKM	four loops, wires at $(\pm 5.30, \pm 5.30)$ and $(\pm 10.25, \pm 10.25)$ mm, length 525 mm, 3000 A; historical peak kick 4.89 mrad at $x = -8.6$ mm ($S = 0.85$); selected $S = 0.95$

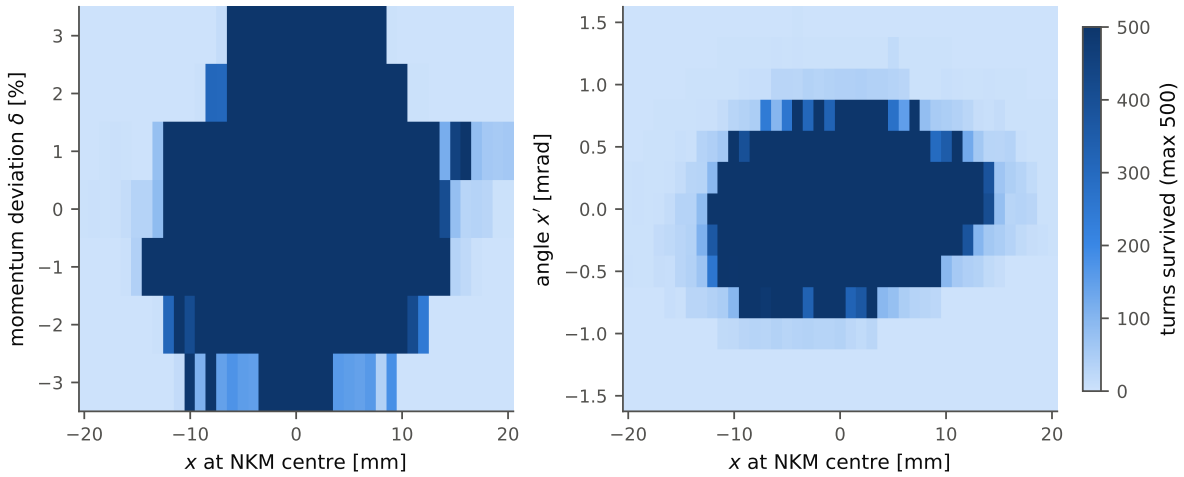


FIG. 1. Deterministic lattice/septum acceptance without the additional chamber profile (no sampling interval), at the NKM centre from native tracking of single particles for 500 turns (1 mm grid in x): turns survived versus x and momentum deviation at $x' = 0$ (left) and versus x and x' at $\delta = 0$ (right).

the fringe and end-connector fields. Its straight-line integrals reproduce both components of the RADIA kick map to 7×10^{-12} rad (4×10^{-10} relative) on 3 657 nodes, so the RADIA calculation and the filament model describe the same field. The tabulated map stores $(\int B_y \, \mathrm{d}s, \int B_x \, \mathrm{d}s)/B\rho$; the curl-free kick pair, $\partial \Delta x'/\partial y = \partial \Delta y'/\partial x$, requires $-\int B_x \, \mathrm{d}s$ for the vertical kick. Using the tabulated field component directly as the vertical kick reverses the vertical response. Physical thin-map comparisons use the coordinate-consistent sign, and the reversed-sign diagnostic is labelled explicitly.

The magnet is tracked in 640 drift–kick–drift slices over ± 0.45 m in canonical AT coordinates, $\mathrm{d}p_x/\mathrm{d}s = PS(B_y - y'B_s)/B\rho$ and $\mathrm{d}p_y/\mathrm{d}s = PS(-B_x + x'B_s)/B\rho$ with slopes $x' = p_x/(1+\delta)$, polarity P and field scale S. The kick acts on canonical momentum and carries no $1/(1+\delta)$ factor; the exit angle changes by 0.12 μrad per 10^{-3} in δ. Slice and span refinement change the exit angle by less than 7×10^{-11} rad. Particles within 1 mm of a conductor are counted as lost. The polarity is chosen so that the kick at $x < 0$ cancels the angle of a beam travelling from the septum towards the axis (figure 2, right).

An independent adaptive DOP853 integration checks the stated paraxial equations at the nominal point, the final angular-window edges (offsets -0.319/0.124 mrad), nonzero vertical position and angle, and $\delta = \pm 0.002$. Relative/absolute solver tolerances are $2 \times 10^{-11}/2 \times$

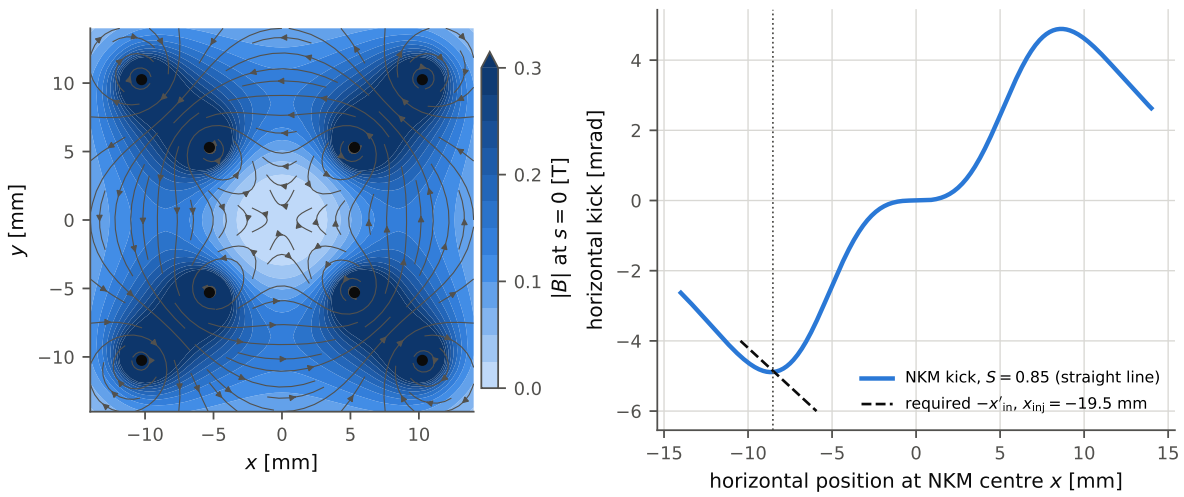


FIG. 2. Deterministic filament-field calculation (no sampling interval). Left: NKM cross-section at $s = 0$ with the wire positions, field lines and $|B|$ of the exact filament field. Right: straight-line horizontal kick at $y = 0$ at the earlier point ($S = 0.85$; the selected point uses $S = 0.95$) and the kick required to cancel the incoming angle of a beam launched from the septum at $x_{\mathrm{inj}} = -19.5\,\mathrm{mm}$; their intersection (dotted) is the thin-kick matched point.

10^{-13}. Maximum 640-slice discrepancies are $4.89 \times 10^{-11}\,\mathrm{m}$ in x, $7.37 \times 10^{-10}\,\mathrm{rad}$ in p_x, $3.11 \times 10^{-11}\,\mathrm{m}$ in y and $8.24 \times 10^{-10}\,\mathrm{rad}$ in p_y. Slice and span checks through 1280 slices and $\pm 0.9\,\mathrm{m}$ are below one percent of a 0.1σ response in either plane; the case-by-case discrepancies are tabulated in the Supplemental Material, Table S1. The field is exact for the filament geometry; tracking remains a validated paraxial approximation.

C. Injected beam, capture and controls

The injected bunch is generated at booster extraction and tracked element by element through the matched BTS line to the septum exit, where it is placed at $(x_{\mathrm{inj}}, x'_{\mathrm{inj}})$. It propagates to the magnet, is tracked through the 3-D NKM once (single-turn pulse, section VIII) and then for 500 turns. *Capture* is the fraction of injected particles alive after the last turn; intervals are 95 % Wilson intervals [24] for particles and bootstrap intervals for error realizations; the statistical definitions, including the ring-cluster bootstrap and the acceptable-loss test, are collected in the Supplemental Material, Sec. S4. Lost particles are classified by upstream transport, septum creation, incoming transport, magnet domain, return septum or

elsewhere in the ring; the categories reconcile with the particle count. Inside the magnet the declared flat chamber is the physical boundary; the 1 mm conductor guard and the ±50 mm field-model domain (the magnet-domain category) are numerical limits that can act only in the fringe span beyond the magnet ends.

The BTS delivers the beam with dispersion $D_x = 80.9\,\mathrm{mm}$ and $D'_x = 47.5\,\mathrm{mrad}$ at the septum exit. The energy spread therefore adds $50\,\mu\mathrm{rad}$ of angular spread to the $57\,\mu\mathrm{rad}$ betatron divergence, and a 0.1 % mean-energy error shifts the centroid angle by $47\,\mu\mathrm{rad}$. A separate transfer-line study finds a dispersion-free exit feasible without changing the line's magnets; its ring capture is not evaluated here.

The *matched* angle is the inner root of zero centroid angle after the magnet. Two controls receive identical particles: a uniform *dipole* kick and a *linearised* kick (with its curl-free vertical partner), both calibrated to the 3-D response of the nominal centroid. This angular calibration does not reproduce every finite-length position–angle correlation, so it cannot uniquely decompose the capture difference into a nonlinear-kick contribution.

III. FINITE-LENGTH AND VERTICAL EFFECTS

For the diagnostic launch $x_{\mathrm{inj}} = -20\,\mathrm{mm}$ and $S = 0.85$, the thin kick map gives a matched angle of 4.870 mrad, the 3-D magnet 4.6475 mrad. To isolate the cause, the map's horizontal profile was distributed over the magnet length and sampled either along the undeflected straight line or along the self-consistent deflected path (figure 3). Straight-path sampling gives 4.762 mrad; the deflected path gives 4.6476 mrad, equal to the 3-D result, and reproduces its response to angle errors to 0.01 mm. Straight-path sampling already shifts the thin-map match, and self-deflection produces an additional shift. Its sagitta is $|K|L/8 = 0.32\,\mathrm{mm}$. Agreement between the deflected-profile and 3-D calculations supports this mechanism in the tested configuration; it does not make longitudinal sampling or end fields universally irrelevant.

The response to septum angle errors is asymmetric. A positive septum angle error moves the trajectory inwards; on the straight line it would cross the kick extremum onto the weaker inner flank, leaving the extra angle under-compensated. The self-deflection instead holds the trajectory on the outer side of the extremum, where the kick is largest (figure 3, centre). A +0.3 mrad error changes the betatron amplitude of the kicked centroid by +1.22 mm with

the thin kick but by -0.12 mm with the 3-D magnet, whereas -0.3 mrad gives $+0.66$ and $+0.60$ mm; the nominal amplitude is larger (9.57 against 9.07 mm). At fixed integrated kick, a longer magnet lowers the sensitivity to positive errors but raises the nominal amplitude, and for $L \geq 0.75$ m no matched solution remains (figure 3, right). Small sagitta relative to the kick-lobe width supports a thin-map approximation; a sliced vector map can resolve the deflection when that ratio is appreciable.

At this diagnostic launch the 3-D magnet defocuses vertically: a centroid at $y = 1$ mm receives 0.25 mrad, against 0.13 mrad from the sign-corrected thin map, because the deflected path samples the field off the straight line. For paired 100 nm rad horizontal-emittance sensitivity bunches at this diagnostic launch, capture is 91.1 % with a reversed vertical-kick sign, 89.7 % with the corrected sign and 81.8 % with the 3-D magnet. Suppressing only the 3-D vertical kick restores 87.8 % (the horizontal-only finite-length model gives 87.4 %), so at that point the vertical defocusing costs about five percentage points. For a curl-free thin vector kick, the linear vertical coefficient is $-\partial \Delta x'/\partial x$, with flank-dependent focusing sign, and vanishes at the kick extremum; at the earlier operating point, where the centroid sat near the extremum, the approximately two-point cost is a 100 nm rad sensitivity result (Supplemental Material, Sec. S3), not the design-beam headline.

The sagitta ratio $\eta = |K|L/(8w)$ compares trajectory displacement with the kick structure. Here K is the peak straight-line kick of the magnet, L its length, and w the full width at half maximum of the negative horizontal kick lobe (9.4 mm for the nominal wires). A trajectory deflected uniformly by K over L is a circular arc whose sagitta relative to its chord is $|K|L/8$; the chord through the kick extremum is the straight-line reference path of a thin map. Figure 4 varies L (0.2–1.0 m), the integrated strength (scales 0.5, 1.0, 1.5) and the transverse wire scale (1.0 and 1.25), giving 30 configurations with $\eta = 0.005$–0.098. Each centroid starts at the straight-line kick extremum with horizontal momentum opposite the kick, $y = 1$ mm and $y' = 0.2$ mrad (both scaled with the wire family), and is tracked by a single thin vector map, a 1280-slice vector map distributed over the wire length, and the exact 1280-slice filament field. Normalizing the exit-momentum error by $|K|$ removes the trivial dependence on kick strength. The thin-map horizontal error then follows $|\Delta p_x|/|K| \simeq 1.95\,\eta^{1.46}$ over all configurations, with a log-rms scatter of 3.4 % (largest deviation 11 %); the two wire families give exponents 1.47 and 1.45. The vertical error organizes less tightly ($\propto \eta^{0.81}$, scatter 7 %), because it also depends on the vertical launch. The

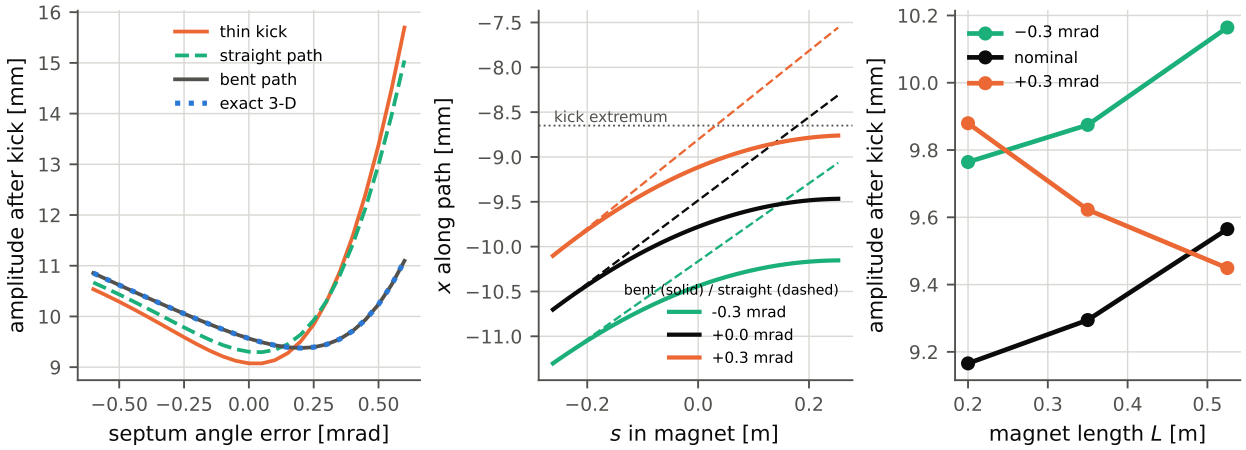


FIG. 3. Deterministic finite-length comparison at the −20 mm diagnostic launch; no sampling interval. Left: betatron amplitude of the kicked centroid at the NKM centre versus septum angle error for the thin kick, the profile sampled on the straight and on the deflected path, and the exact 3-D field, each at its own matched angle. Centre: centroid path inside the magnet for angle errors of −0.3, 0 and +0.3 mrad (solid: deflected; dashed: straight) relative to the kick extremum. Right: deflected-path model versus magnet length at fixed integrated kick.

sliced vector map stays below 2×10^{-5} of $|K|$ in both planes.

To relate this to injection, we declare an accuracy allowance of one tenth of the 80 % angular-window width (0.044 mrad), the same convention as the pulse allowance of section VIII. At the nominal kick the fitted thin-map error reaches it at $\eta^* \simeq 0.025$; for the nominal magnet ($\eta = 0.034$) the per-pass thin-map error is 155 % of the allowance ($|\Delta p_x| = 0.069$ mrad, and $|\Delta p_y| = 0.129$ mrad at the stated vertical launch), so with the angular window of the selected operating point a single thin map is not an adequate description.

This launch is a diagnostic recipe. Along the matched trajectory of the selected point, which crosses the steep inner flank of the kick, the thin-map horizontal error is 161 μrad, 3.7 times the allowance; near the kick extremum (the earlier and extremum-centred points of section VII) it is about 47 μrad. The 1280-slice map agrees with the exact field to better than 9×10^{-8} rad on all three trajectories. The empirical η fit describes deterministic comparisons for the tested family and launch recipe; it establishes neither a universal law nor a universal capture criterion. A sliced vector map retains self-deflection and the physical vertical kick; the local focusing sign changes between flanks of the kick extremum. Reversing the vertical-

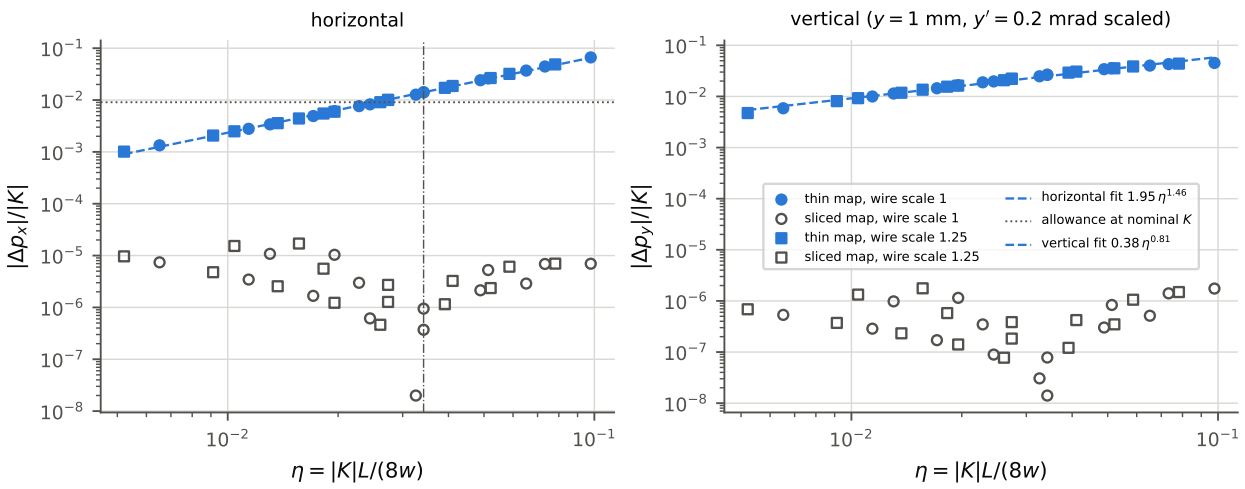


FIG. 4. Exit-momentum errors of a single thin vector map (filled) and a 1280-slice vector map (open) against the exact filament field, normalized by the peak kick $|K|$, versus η for five lengths, three integrated strengths and two transverse wire families (circles, squares). Left: horizontal, with the power-law fit, the declared allowance at the nominal kick (dotted) and the nominal magnet (dash-dotted). Right: vertical, for the stated vertical launch. Deterministic comparisons; no sampling interval.

kick sign introduces an input-convention error; it does not reveal an inherent failure of kick maps. Capture controls throughout are paired model comparisons on identical particles, not independently optimized injection systems.

IV. OPERATING WINDOW

At the final point and reference chamber, the native ring captures 98.4 % at 500 turns, versus 99.5 % for an affine element-map reference on identical particles. The reference retains element lengths and explicit aperture checks at physical boundaries, and linearizes about the six-dimensional closed orbit with RF and radiation. A small-perturbation one-turn check agrees within 1.36×10^{-14} in the corresponding metre/radian/dimensionless coordinates. Because the affine reference omits nonlinear dynamics, all capture conclusions use native tracking.

Nominal native survival at 4000 turns is 97.8 % (95 % particle Wilson interval 96.7–98.5 %), a loss of 0.6 percentage points after turn 500. The 500-turn prefix reproduces the paired chamber-study particle masks exactly. Fresh bunches at the interpolated lower and

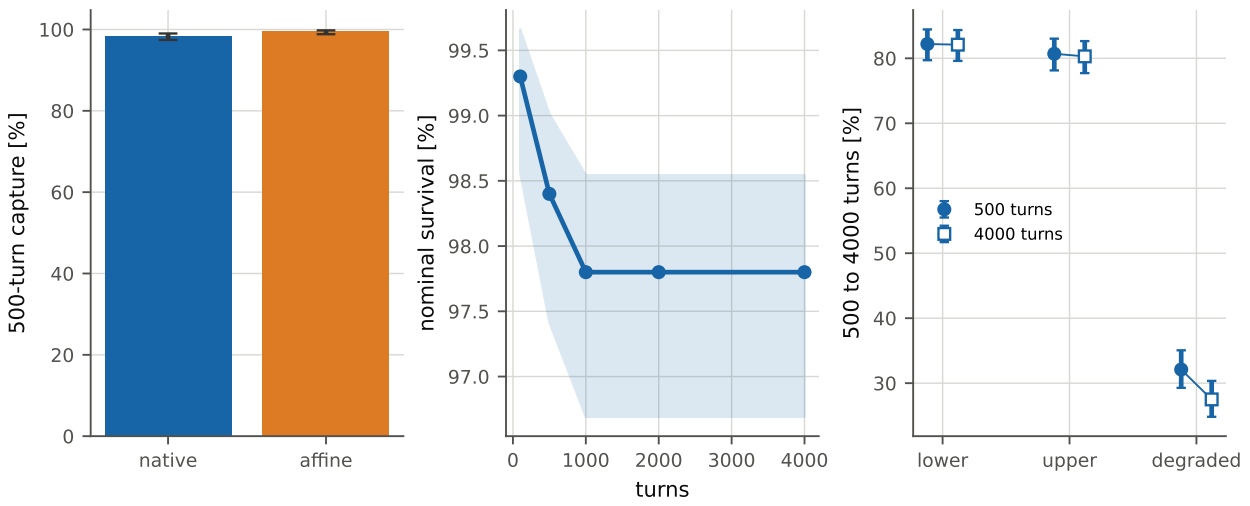


FIG. 5. Final-point validation with the 3-D NKM, design beam and reference chamber. Left: native and affine-element backends at 500 turns on the same two 500-particle bunches (seeds 1001/1002; 95 % particle Wilson bars). Centre: native nominal survival with turn count (same particles; Wilson band). Right: 500-to-4000-turn survival for interpolated lower/upper angular-edge candidates and corrected ring 5501, each with two fresh 500-particle bunches (seeds 2001/2002); circles/squares denote 500/4000 turns, with 95 % Wilson bars. Intervals are conditional on the selected models and ring, not machine-population uncertainty.

upper angular-edge candidates give 82.2/80.7 % at 500 turns and 82.1/80.3 % at 4000 turns (95 % intervals 79.6–84.4 % and 77.7–82.6 %). The corrected ring with the lowest validated capture (5501) gives 32.1 % at 500 turns and 27.5 % at 4000 turns (24.8–30.3 %). The stated window is therefore conditional on 500-turn survival. It does not establish infinite-time acceptance or guarantee a 4000-turn contour at 80 %.

The operating point is selected for the design beam on the corrected-ring population (section V) with the declared chamber, by a rule fixed before the search. The injected centroid keeps $3\,\sigma_x$ of the design beam ($\sigma_x = 159\,\mu$m at the septum) from the outer face of the blade; x_{inj} is set at that limit and 0.25, 0.5 and 1.0 mm further out, S ranges over 0.80–1.00, and x'_{inj} is the inner matched root of the 3-D magnet. Each of the 40 candidates, for the 2 mm study blade and the 1 mm KLS thin septum, is scored by the tenth percentile, then the mean, of capture over the ten corrected rings without injection errors (100 particles per ring). For the 2 mm blade the rule selects $x_{\mathrm{inj}} = -19.48$ mm, $S = 0.95$ and $x'_{\mathrm{inj}} = 5.319$ mrad, which places the kicked centroid at -7.44 mm instead of -8.52 mm. On fresh bunches (two of

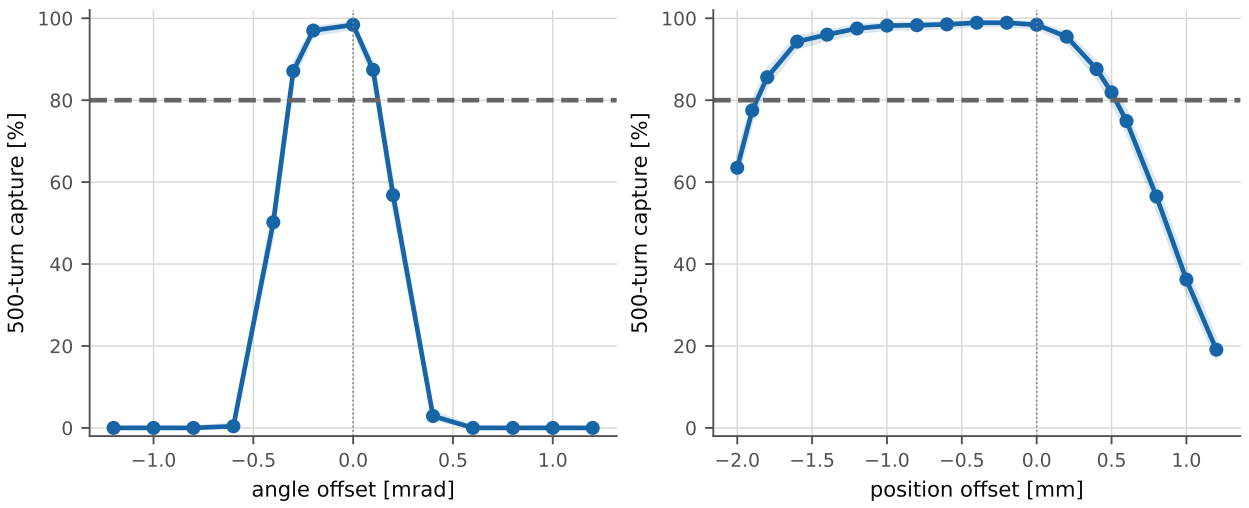


FIG. 6. Design-beam operating window with the reference chamber and fixed NKM settings. Left/right: capture versus septum-exit angle/position offset, 500 native turns, two 500-particle bunches (seeds 1001/1002) per point. Bands are conditional 95 % particle Wilson intervals; the dashed line marks 80 %. Crossing cells have 0.1 mrad/mm spacing after refinement.

500 particles per ring) it captures 98.4 % on the ideal ring and 78.0 % on average over the corrected rings (tenth percentile 65.7 %), against 97.0, 68.9 and 43.3 % for the earlier point ($x_{\mathrm{inj}} = -19.5\,\mathrm{mm}$, $S = 0.85$), which had been chosen with the 100/10 nm rad sensitivity beam. The 1 mm blade gives nearly the same result (78.8 %). The gain comes from the smaller injected amplitude, obtained through kick strength and angle; moving x_{inj} to the clearance limit at fixed S does not help, and the selection ranking is flat within its noise.

We estimate the connected operating window above 80 % capture around the selected point on the ideal ring. Linear interpolation gives angle offsets from -0.32 to 0.12 mrad (width 0.44 mrad), and position offsets span -1.87 to 0.53 mm (width 2.40 mm). Each value uses seeds 1001/1002 and 500 particles per seed. A fixed 0.2 mm/mrad grid is extended if a boundary remains above threshold, with a maximum offset of 3; crossing cells are refined once to 0.1 spacing and linearly interpolated. Grid resolution is not a confidence interval for a crossing. The long-turn edge checks use fresh particles and do not redefine the selected 500-turn window. Compared with the earlier point (0.76 mrad and 1.99 mm), the selected point trades angular width for position width; the robustness study below shows which matters under the error budget.

V. SURVIVAL, ACCEPTANCE AND SENSITIVITY

The earlier larger-emittance sensitivity case (100/10 nm rad, septum aperture only), including its 4000-turn survival and the paired vertical-kick diagnostic, is documented in the Supplemental Material, Sec. S3; its numbers are not design-beam results.

A separate corrected-ring scenario applies correlated 30 μm girder offsets, 30 μm independent magnet offsets, 100 μrad rolls, gradient/sextupole errors and skew-quadrupole components, all explicit sensitivity assumptions, to ten fixed seeds (5501–5510). The correction chain follows commissioning practice within the hardware of the supplied lattice. Each BPM reads the orbit relative to its adjacent quadrupole centre, as after beam-based alignment, with a 10 μm rms residual. The orbit is corrected with the 120 HVC correctors by truncated SVD, iterated until the rms improves by less than 1 % per step. The β-beating and horizontal dispersion at the BPMs are then corrected with individual quadrupole gradients (Gauss–Newton on a fixed Jacobian, gradient changes bounded to 1 % per step), followed by bounded Q11/Q12 tune correction and a final orbit correction. The lattice has no skew correctors, so coupling is not corrected. The rms β-beat at the BPMs falls from 4.3–10.5 % to at most 2.0 %, and the closed orbit at the NKM stays within 56 μm. The orbit relative to the quadrupole references converges to 69–83 μm horizontally and 49–63 μm vertically, with unsaturated correctors; this floor reflects the 120-corrector system of the file, not a stopping rule. Final fractional-tune residuals are below 3.5×10^{-5}. The first unstable horizontal launch lies between -11.0 and -9.5 mm for all ten rings (figure 7); ring 5501, whose first unstable launch was -7.5 mm with orbit correction alone, reaches -10.5 mm once its optics are corrected. These ten rings define the simulated scenario; their results do not predict commissioned-machine performance.

The reference arc half-apertures are 12 mm horizontally and 9 mm vertically [25, 26]. The centre-bend vertical half-aperture is 4.87 mm. An inscribed ellipse represents the unspecified octagonal corners. Inside the magnet the beam passes between the conductors, so the NKM vacuum chamber must be flat and lie below the inner wires at $|y| = 5.30$ mm. Following the allowance of 1.5 mm for chamber wall plus conductor radius used for the eight-conductor ALS kicker [18], the assumed chamber is 16×3.8 mm in half-width and half-height over the magnet length; its 32 mm full width matches the ALS prototype. In units of $\sqrt{\beta_y}$ its vertical half-aperture, 1.61 mm m$^{-1/2}$, is smaller than that of the arcs (1.81) and centre bends (2.64),

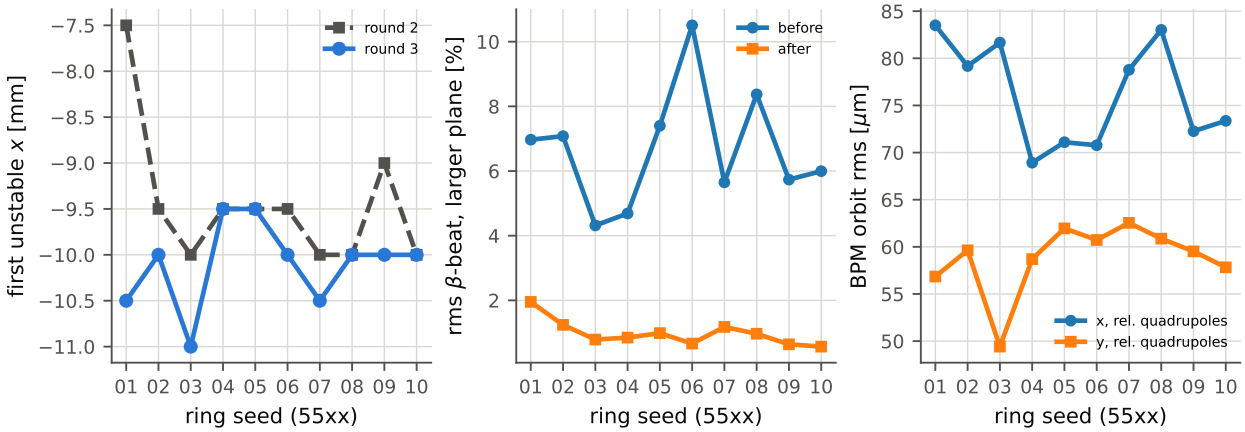


FIG. 7. Corrected-ring scenario, ten fixed seeds. Left: first unstable horizontal launch at the NKM centre (500 native turns, RF and radiation, septum aperture only) after orbit correction alone (round-2 rings, squares) and after the full correction chain (circles). Centre: larger-plane rms β-beat at the BPMs before and after optics correction. Right: BPM orbit rms relative to the quadrupole references after the final correction. Deterministic quantities; no sampling interval.

so with open IVUs the NKM chamber is the tightest vertical restriction of the ring. Outside the magnet the injection straight keeps an assumed 30×15 mm rectangle. The chamber acts on the injection pass, through the sliced kicker, and on every later turn, as the aperture of the two magnet halves in the ring; the thin controls are checked at the magnet centre.

On the ideal ring with the design beam, evaluated at the earlier operating point ($x_{\mathrm{inj}} = -19.5$ mm, $S = 0.85$), it is not limiting: nominal capture is 97.0 % (95.7–97.9 %) with the reference chamber and equal for half-heights from 3.0 to 4.3 mm and without the chamber, and no particle is lost in the magnet; the injected vertical beam size, about 60 μm, leaves a large margin. On corrected rings, later vertical losses can occur at this chamber. Opening it alone does not remove the slow-loss limitation (section VII); the interception boundary and the dynamics that drive the loss are distinct.

Nineteen identified ordinary straights receive 3 m IVU segments [26]; their assignment is a scenario, not a surveyed map. The reference open-IVU half-gap equals the arc half-height, 9 mm; the impedance model of Ref. [25] uses a 16 mm open gap (8 mm half-gap). With $\beta_y \leq 3.6$ m in those straights, either value gives a normalized vertical aperture of at least $4.2\,\mathrm{mm\,m^{-1/2}}$, far above that of the arcs (1.81), so the choice does not affect capture. A 2.5 mm closed magnetic half-gap is a stress case and does not account for a liner. The

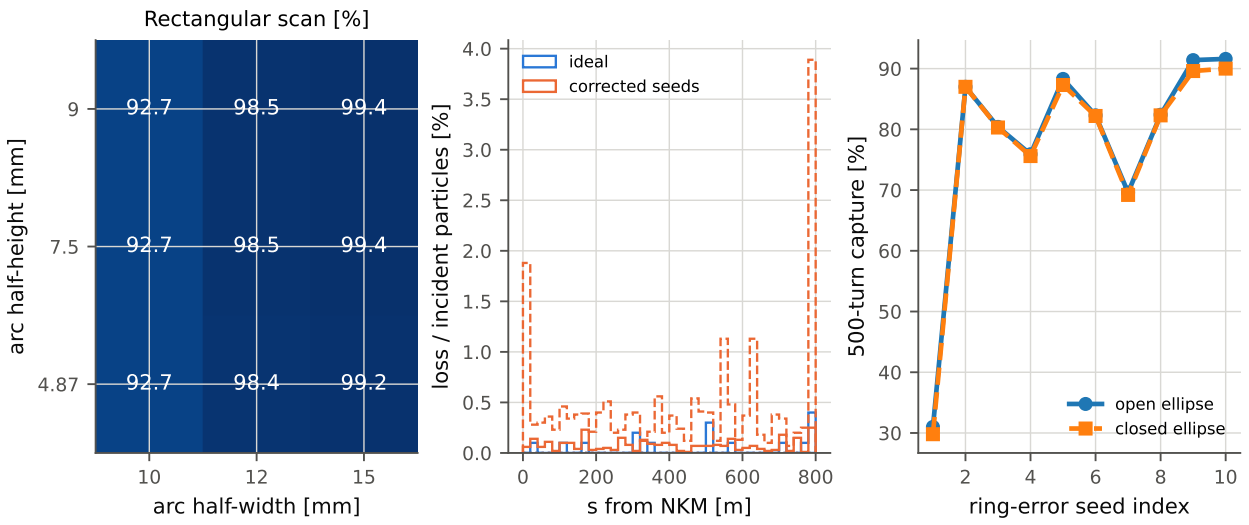


FIG. 8. Declared chamber with the design booster beam, 500 native turns and two seeds of 500 particles per ring/profile/model. Left: ideal-ring capture versus rectangular arc half-apertures, with other region limits retained. Centre: ring losses per incident particle in 20 m bins using element-entrance positions from the NKM centre; solid/dashed/dotted denote horizontal/vertical/dynamical losses. For elliptic apertures the larger normalized coordinate defines the plane. Right: capture for each corrected ring with open and closed IVUs. Points are conditional sample fractions; the loss histograms and finite ring-seed variation have no assigned confidence band.

separate incoming transport retains its source lattice and septum creation cut; its chamber has not been surveyed for this alternative injection scheme.

For the 7.67 nm rad design booster beam, the reference chamber gives 98.4 % capture at 500 turns (95 % particle Wilson interval 97.4–99.0 %; two 500-particle bunches, seeds 1001/1002). The paired dipole gives 100.0 %; the NKM-minus-dipole difference is -1.6 percentage points (95 % paired-particle bootstrap interval -2.4 to -0.9; 10 000 draws, seed 1729). In rectangular arc scans, half-widths 10, 12 and 15 mm give 92.7, 98.5 and 99.4 % at each tested half-height (4.87, 7.5, 9 mm). Thus 10 mm is the smallest tested rectangular-grid width exceeding 80 % in this ideal-ring sample, while 12 mm retains the reference result; neither is a continuous minimum or a machine tolerance. Across the ten corrected rings, mean capture is 78.0 %, median 82.3 %, tenth percentile 65.7 %, and range 31.0–91.6 %. These are different descriptors of a finite scenario population.

VI. STORED-BEAM TRANSPARENCY

The kick of the four-loop magnet is odd in x on the midplane and its near-axis expansion has a leading cubic term, $\Delta x' \simeq c_3 x^3$ with $c_3 = 2.6 \times 10^4\,\mathrm{rad\,m^{-3}}$ at the selected $S = 0.95$. For an aligned kicker and a centred, symmetric Gaussian beam the ideal centroid kick therefore vanishes exactly. This does not imply unchanged finite-beam moments: deterministic quadrature gives fractional single-pass changes of the centred projected emittance, $\Delta\varepsilon/\varepsilon$, of 5.6×10^{-7} horizontally and 2.0×10^{-7} vertically, from the kick spread across the beam alone. Orders 20, 40 and 60 per axis agree at the reported precision. We evaluate the response to an offset between kicker and stored orbit, together with the emittance change, from the exact field by deterministic Gauss–Hermite quadrature (figure 9, right). Offsets of 0.1, 0.2 and 0.5 mm give centroid amplitudes A_x of 0.018, 0.12 and 1.72 of the rms size, and the spread of the kick across the beam increases the emittance about the displaced centroid by 1.2×10^{-4}, 1.5×10^{-3} and 5.3×10^{-2}; a 0.2 mm vertical offset gives 0.19 of the vertical rms size. Once the centroid oscillation has also filamented, the rms emittance grows by the factor $1 + A_x^2/2 + \Delta\varepsilon_x/\varepsilon_x$, i.e. 1.008 at 0.2 mm and 2.52 at 0.5 mm.

Native tracking uses two seeds (11 and 12), each with 300 paired particles, with radiation on. The maximum centroid difference is measured at the ends of turns 1–50, and projected emittance at the end of turn 1000: $A_x/\sigma_x = 0.11$ and 1.65, and mean paired emittance ratios 1.01 and 2.55. These tracked observables need not equal a fully filamented analytic estimate.

In this ideal geometry and at the assumed emittance, an offset of 0.2 mm keeps the stored-beam disturbance near a tenth of its size, slightly above the $0.1\,\sigma$ criterion at the selected strength, whereas 0.5 mm is not transparent. These responses, the tracking and the tolerances below are evaluated at the selected $S = 0.95$; the single-pass kicks are linear in S, so the values at the earlier $S = 0.85$ are smaller by the factor 1/1.12. The bicubic interpolation of the map agrees with the exact field to 1×10^{-10} rad near the axis, whereas bilinear interpolation errs by 1×10^{-6} rad and overstates the 0.1 mm response nineteen-fold (0.33). The calibrated dipole control would displace the stored beam by about 2495 rms sizes in a linear estimate; this hypothetical figure only illustrates why a uniform kicker cannot be used for top-up.

The assumed vertical-to-horizontal emittance ratio is 0.10, as supplied by the author; for

a fixed vertical dipole kick, $A_y/\sigma_y \propto \varepsilon_y^{-1/2}$. Individual wire displacements and independent loop-current errors break the cancellation; the per-wire and per-loop sensitivities are tabulated in the Supplemental Material (Tables S2 and S3). The one-error-at-a-time wire limits range from 0.26 to 1.82 μm for both centroid amplitudes to remain below $0.1\,\sigma$. Independent loop currents require relative balance of about 1.0×10^{-4}. In series powering, a common current error preserves dipole cancellation provided the geometry remains symmetric; geometry errors still matter.

We also evaluate simultaneous errors. Displacing all eight wires by independent Gaussian offsets (4000 draws per case, series powering) gives a median stored-beam amplitude of 2.9 and 5.4 rms sizes horizontally and vertically for 10 μm rms placement, 7.2 and 13.3 for 25 μm and 14.8 and 25.9 for 50 μm; independent loop-current errors of 10^{-4}–10^{-3} change these by at most 3 %. Even at 10 μm only 0.03 % of the draws keep both planes below $0.1\,\sigma$. Geometric symmetry therefore does not provide transparency for a magnet built to realistic tolerances: the static on-axis dipole residual must be measured and compensated, for example by a trim winding or a pulsed corrector, as at SOLEIL [12]. The supplemental tables give the per-wire sensitivities that compensation must address. Designing that compensation is outside this study.

The paired controls (figure 9, left) compare the models at the earlier point: the linearised and dipole kicks capture 97.8 and 97.0 %, the paired NKM-minus-dipole difference is -4.1 points (95 % paired-bunch bootstrap interval -4.7 to -3.2), conditional on this larger-emittance sample; with the kicker off no particle survives.

The septum also acts on the stored beam. At the septum the horizontal beta function is 15.9 m, so a stored-beam centroid amplitude of $0.1\,\sigma_x$ corresponds to a dipole kick of 197 nrad, i.e. a uniform leakage field of 0.066 G over the 0.4 m thin septum. The published baseline value, about 3.3 G at the bumped orbit 5 mm from the blade [23], is 50 times larger. In the NKM scheme the stored orbit is about 16 mm from the blade, where the leakage is lower but has not been published; the requirement is therefore a leakage of order 0.07 G at the stored orbit, by distance or shielding. The shielded thick-septum value, 0.3 ± 0.1 G over 1.7 m per magnet, would by the same measure exceed the allowance unless those magnets act only on the transfer line. Septum leakage may therefore set the system-level transparency limit even when the NKM contribution is cancelled.

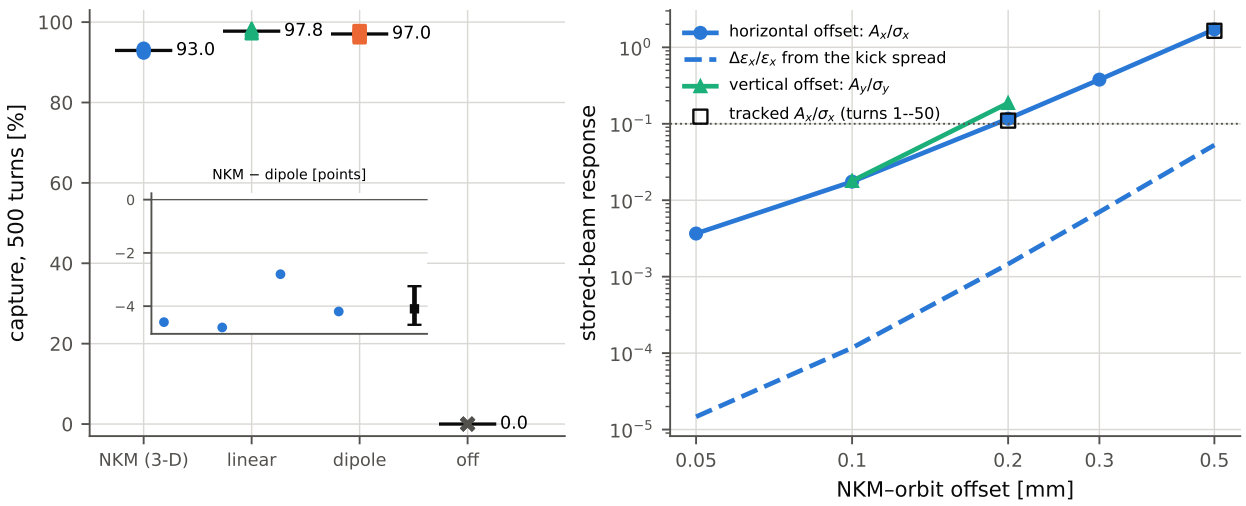


FIG. 9. Left: 100/10 nm rad sensitivity case with the septum aperture and no additional chamber; four fresh 500-particle bunches at the earlier point ($x_{\mathrm{inj}} = -19.5\,\mathrm{mm}$, $S = 0.85$) for the 3-D NKM, the calibrated linearised and dipole controls and the kicker off (markers: bunches; bars: mean). The inset shows per-bunch NKM-minus-dipole differences (dots), their mean (square) and a paired-bunch bootstrap 95 % interval (10 000 resamples, seed 1729; only four bunches). Right: deterministic centroid amplitude and filamented emittance growth versus NKM-to-orbit offset. Open squares are the mean of two 300-particle tracked seeds, using maximum centroid differences over ends of turns 1–50; they are not 1000-turn amplitudes. Emittance ratios in the text use the end of turn 1000. No sampling interval is assigned to deterministic quadrature.

VII. ROBUSTNESS AND TOLERANCES

A. Error budget

Table III defines the final untruncated Gaussian rms budget. Booster horizontal emittance 7.67 nm rad and energy spread 0.106 % are design values [27]; extraction Twiss, bunch length and the 0.10 vertical-emittance ratio are assumptions. No public extraction-orbit or septum-current stability was supplied, so the corresponding errors remain sensitivity parameters. Capture robustness is evaluated separately from exit Twiss mismatch.

The published bump-kicker reproducibility is not treated as an NKM or septum specification. At the lattice RF frequency, phase error shifts the injected longitudinal coordinate by $c\Delta\phi/(2\pi f_{\mathrm{RF}})$. Independent arrival relative to NKM firing scales the 5.3 μs half-sine field at

TABLE III. Final rms error budget. S: static machine draw; J: shot draw. Unsourced entries are explicit sensitivity assumptions with the tested effect stated. Corrected-ring errors are a separate finite scenario population.

Term	RMS	Class	Source or reason
Booster centroid x/x'	0.5 mm / 0.2 mrad	J	Extraction stability unavailable; scan scales 0–1
Booster beta mismatch	5%	S	Assumed extraction-optics sensitivity
Emittance / mean energy	10% / 0.1%	J	Assumed fluctuations about design beam
Energy-spread variation	10^{-4} absolute	J	Assumed variation of within-bunch spread
BTS quadrupole gradient	0.1%	S	Assumed focusing error
BTS quadrupole x/y offset	100 μm each	S	Assumed alignment error
BTS quadrupole roll	0.5 mrad	S	Assumed coupling error
BTS dipole field	0.05%	S	Assumed rigidity/trajectory sensitivity
Septum position	0.1 mm	S	Assumed blade geometry error
NKM / residual orbit offset	0.2 mm each	S	Assumed relative alignment sensitivity
Common NKM amplitude	0.5%	J	Assumed pulser fluctuation, not a measured specification
Thin-septum strength	0.1% relative	J	Assumed; nominal bend 10 mrad [23]
RF phase	0.001 rad	J	Assumed injection-phase sensitivity
Arrival relative to NKM firing	2 ns	J	Assumed peak-phase pulse timing

peak phase and does not add another longitudinal displacement. The separate thin-septum error adds canonical momentum $(1+\delta)\,0.01\,\epsilon_{\mathrm{sept}}$ in radians. No equivalent transverse timing-kick term is added. Longitudinal quadrupole placement, supply quantization and a scalar ring-beta perturbation are not modelled and are not counted as applied errors.

B. Combined errors and tolerance curves

Three seeds (42, 123, 777) generate 150 joint machine/shot realizations, each with 200 Gaussian particles and 500 native turns. Each realization independently draws one of the ten corrected rings with replacement, plus the errors in table III. The bootstrap unit is the joint realization, conditional on that finite ring population; it is not an estimate over all possible commissioning errors or repeated shots on one machine. A ring-cluster bootstrap and leave-one-ring-out means show the dependence on the ten rings (Supplemental Material, Sec. S4).

Ideal static steering transports the zero-input reference through the perturbed BTS, giving (r_x, r_{p_x}), then translates the bunch at the septum by $(x, p_x) \mapsto (x - r_x, p_x - r_{p_x})$. The other coordinates are unchanged. This removes no shot jitter, vertical trajectory error or optics distortion, and does not model a fitted orbit-correction system. A declared sensitivity applies the same ideal translation in both planes (section VII C).

The 3-D NKM mean is 16.1 % (95 % interval 12.0–20.4 %; ring-cluster interval 10.9–21.5 %, leave-one-ring-out range 14.9–17.4 %), median 0.0 %, tenth percentile 0.0 %, and fraction of realizations at or above 80 % is 5.3 %. The paired dipole mean is 44.1 % (38.7–49.6 %). The NKM-minus-dipole mean difference is -28.0 percentage points (-32.5 to -23.5). These controls compare models on identical particles and errors; they are not separately optimized injection systems. The capture distribution is bimodal: 51 % of the 150 realizations capture nothing with the NKM, against 10 % with the dipole, and the means over realizations with nonzero capture are 33.0 and 49.0 %.

The jitter scan retains the same 50 seed-42 realizations, particle seeds and ring identities while scaling only booster position and angle jitter; table IV gives both models. Quarter jitter corresponds to 0.125 mm and 0.05 mrad; its paired NKM difference is -5.7 points (-9.4 to -2.5). The whole interval lies below the declared one-point acceptable-loss margin, so even a quarter of the assumed jitter fails the noninferiority test; the lower endpoint of a

TABLE IV. Paired jitter sensitivity for both models, 50 joint realizations per row (seed 42, horizontal-only steering). Means and zero-capture fractions are percentages; differences from zero jitter and the NKM-minus-dipole difference are percentage points with paired 95 % realization-bootstrap intervals (10 000 resamples, seed 1729).

	NKM			Dipole			
Scale	Mean	Zero	Difference	Mean	Zero	Difference	NKM−dipole
0	40.1	18	—	68.5	2	—	-28.3 [-35.5, -21.1]
0.25	34.4	28	-5.7 [-9.4, -2.5]	66.9	0	-1.5 [-3.0, -0.2]	-32.5 [-40.2, -24.8]
0.5	28.8	34	-11.4 [-17.0, -6.0]	61.6	0	-6.9 [-11.1, -3.4]	-32.8 [-40.9, -24.9]
0.75	24.0	44	-16.2 [-22.8, -9.6]	54.9	6	-13.5 [-20.2, -7.6]	-31.0 [-39.6, -22.6]
1	19.5	46	-20.7 [-28.2, -13.6]	48.2	12	-20.2 [-28.6, -12.7]	-28.8 [-37.7, -20.1]
position only	31.2	30	-8.9 [-13.6, -4.6]	65.8	0	-2.6 [-5.4, -0.1]	-34.6 [-42.2, -27.1]
angle only	23.3	40	-16.8 [-24.0, -10.2]	50.5	10	-18.0 [-25.8, -11.0]	-27.1 [-35.4, -19.1]

central 95 % interval, used as the test bound, is a conservative 97.5 % one-sided bound. Full jitter costs both kickers about 20 points and angle-only jitter costs the dipole at least as much as the NKM, so under the production steering booster jitter is not the NKM-specific loss; position-only jitter is the more NKM-specific component. Mean, zero-capture fraction and paired differences answer different questions; no 80 % design specification follows from a nominal ideal-ring value.

An exploratory association analysis uses signed errors and a physically constructed predictor, the horizontal betatron amplitude of the centroid just after each kicker, computed from each realization's errors (150 full-budget realizations; Spearman rank correlation with Holm adjustment within each response; Supplemental Material, Sec. S4). Capture follows each kicker's own post-kick amplitude ($\rho = -0.60$ for the NKM and -0.70 for the dipole). The signed NKM-to-orbit offset is associated with NKM capture only ($\rho = +0.26$, $p_{\mathrm{Holm}} = 0.011$; dipole $+0.17$), a one-sided effect that a magnitude-based analysis cannot detect, and the signed booster position with the NKM-minus-dipole difference ($\rho = +0.28$, $p_{\mathrm{Holm}} = 0.005$). These are associations; the attribution in section VII C uses paired ablation.

For capture-based tuning, each of the 50 seed-42 static machines receives 100 fresh training particles with shot fluctuations disabled. The knobs are x_{inj}, bounded above by the

septum-clearance limit and 1.5 mm below it, an angle trim of ±0.5 mrad about the re-matched angle of the current (x_{inj}, S), and S in 0.75–1.05; repeated strict-improvement coordinate sweeps (steps 0.25 mm, 0.1 mrad and 0.025, at most four sweeps; median 13 evaluations) re-match the angle at every trial. Independent held-out 200-particle samples then restore the joint shot errors at zero and quarter booster jitter. The tuned-minus-untuned gains are +11.9 points (+8.0 to +16.3) and +11.0 points (+7.0 to +15.4), respectively, with paired 95 % realization-bootstrap intervals. The corresponding tuned means are 52.1 and 45.4 %; fractions of realizations above 80 % are 30.0 and 24.0 %. 28 % of the machines end on the outer x_{inj} bound, i.e. they prefer a larger injection offset than the declared range allows; no machine ends on the angle or field-scale bounds.

Training capture has one-percentage-point counting resolution. No BPM calibration, resolution, charge-monitor noise or actuator-error model is included, so this reduced ideal-observable search quantifies recovery without demonstrating operational beam-based feedback. Held-out results are never used to choose knobs, and the dipole control is not retuned.

Table V collects the capture of the selected point as the conditions accumulate; the populations differ between rows (stated in the first column), so differences between rows are not paired effects. Without injection errors the corrected rings reduce capture almost equally for the NKM and the dipole (78.0 and 82.2 %); the ring-to-ring spread is common to both kickers. With ten rings, residual chromaticities spanning only 5.9–6.9 and unadjusted tests, its associations with chromaticity ($\rho = -0.68$) and β-beat (-0.58) do not identify a mechanism. Nor does the margin between the first unstable launch and the injected centroid amplitude organize capture ($\rho = -0.23$ over 400 candidate–ring pairs); a logistic fit of capture against the margin is degenerate, its width ending on the 5 mm fit bound, so this statement rests on ρ alone. The loss that distinguishes the NKM appears only with injection errors and is analysed next.

C. Attribution of the NKM-specific loss

The NKM-specific loss is present without booster jitter: the joint budget at zero jitter leaves the NKM at 40.1 % against 68.5 % for the dipole. A paired ablation on the same 50 seed-42 realizations, particles and corrected rings attributes it (table VI). The error terms of table III are grouped into seven physical classes; each class is switched on alone and

TABLE V. Capture of the selected point as conditions accumulate (500 turns unless stated). Ideal-ring rows: fresh 2×500-particle bunches; corrected-ring row: mean over ten rings, 2×500 particles each; joint-error rows: means over 50 (jitter rows, seed 42) or 150 joint realizations of 200 particles. Rows use different samples and are not paired.

Condition	NKM [%]	Dipole [%]
Ideal ring, no errors, 500 turns	98.4	100.0
Ideal ring, no errors, 4000 turns	97.8	—
Corrected rings, no injection errors	78.0	82.2
Joint errors, no booster jitter	40.1	68.5
Joint errors, quarter booster jitter	34.4	66.9
Joint errors, full budget (150 draws)	16.1	44.1
Joint errors, no booster jitter, tuned	52.1	—

switched off from the full budget, with every other random draw unchanged. With all classes off, capture is 79.5 and 84.5 % (difference -5.0 points). The BTS magnet errors alone change NKM capture by -39.2 points (-48.0 to -30.5) and dipole capture by -14.5 points, and omitting them from the full budget restores 76.6 and 82.6 %. Every other class changes either model by at most 1.6 points. The classes add nearly linearly (non-additivity +3.5 points for the NKM, +0.2 for the dipole).

The BTS magnet errors act through the vertical plane. The modelled steering removes only the horizontal static trajectory, so quadrupole offsets and rolls leave a vertical centroid error of 0.90 mm and 77 μrad rms at the septum exit; within that ensemble NKM capture falls with $|y|$ ($\rho = -0.88$, dipole -0.52; exploratory). A vertically displaced centroid receives a vertical kick proportional to the local horizontal kick slope (section III), which is steep at the selected point and absent for the uniform kick. The loss locations support this interpretation.

In the first NKM zero-capture realization on each of 8 rings, 1 491 of the 1 500 NKM losses occur on the first turn, 955 of them in the vertical plane, spread around the ring, and none inside the magnet; the dipole loses the same realizations over many turns. A declared sensitivity removes the static trajectory in both planes; this is still an ideal translation and does not change the transfer line. Without booster jitter it raises capture to 76.6 and 82.6 %

TABLE VI. Paired error-class ablation without booster jitter (50 seed-42 realizations, 200 particles, 500 turns, horizontal-only steering). Alone: the class switched on with all others off, minus all-off. Omitted: the full zero-jitter budget minus the budget without the class. Percentage points; the NKM-minus-dipole differences carry paired 95 % realization-bootstrap intervals.

	Alone			Omitted		
Class	NKM	Dipole	NKM−dipole	NKM	Dipole	NKM−dipole
BTS magnets	-39.2	-14.5	-24.7 [-31.7, -17.7]	-36.4	-14.1	-22.3 [-28.7, -16.1]
Beam shape	-0.2	+0.1	-0.2 [-0.8, +0.4]	+0.2	+0.0	+0.1 [-0.7, +1.0]
Mean energy	-1.0	-0.7	-0.3 [-1.9, +1.1]	+0.3	+0.1	+0.2 [-2.0, +2.7]
NKM amplitude/timing	-0.6	-0.2	-0.4 [-1.1, +0.2]	+0.1	-0.6	+0.7 [-0.1, +1.6]
NKM/orbit alignment	-1.6	-0.4	-1.2 [-2.6, +0.2]	-0.3	-0.5	+0.2 [-2.2, +2.7]
Septum	+0.1	-0.4	+0.5 [-0.3, +1.2]	+0.4	-0.1	+0.5 [-0.3, +1.3]
RF phase/arrival	-0.2	+0.0	-0.2 [-0.7, +0.2]	+0.5	+0.2	+0.2 [-0.4, +0.9]

(paired gains +36.4 and +14.2 points), the value with the BTS magnet errors omitted, and the difference shrinks to -6.1 points. With the full budget it raises capture to 33.9 and 57.2 %, but the difference remains -23.3 points.

Once the vertical error is removed, booster jitter changes NKM capture by -42.7 points and dipole capture by -25.4 (difference -17.2, interval -25.4 to -9.2), whereas with horizontal-only steering it costs both about 20 points. The static vertical error and booster jitter therefore remove largely the same NKM shots. Which contribution appears NKM-specific depends on which error is corrected.

Splitting the jitter (a declared extension on the same realizations) shows that both components are then NKM-specific. Position-only jitter changes NKM and dipole capture by -17.4 and -2.6 points (difference -14.7, interval -21.8 to -8.2); angle-only jitter by -33.7 and -20.5 points (difference -13.2, interval -21.2 to -5.8). The NKM kick depends on where the centroid crosses the magnet, so a position error leaves the kicker as an angle error; the uniform kick has no such dependence, consistent with its small position-jitter cost.

The deficit also depends on the operating point (table VII). The earlier point and a point whose matched trajectory is centred on the kick extremum were evaluated on the same realizations without and with booster jitter. NKM capture does not change resolvably (full-

TABLE VII. Operating points evaluated on the same 50 seed-42 realizations, particles and corrected rings: mean capture [%] without (zero) and with (full) booster jitter. x_c is the matched centroid at the NKM centre; the dipole control is recalibrated at each point.

Point	x_{inj} [mm]	S	x_c [mm]	NKM, zero	Dipole, zero	NKM, full	Dipole, full
current	-19.48	0.95	-7.44	40.1	68.5	19.5	48.2
earlier	-19.50	0.85	-8.52	39.1	45.7	22.4	30.4
extremum-centred	-20.75	0.95	-8.50	37.8	46.0	21.6	31.0

budget differences +2.9 and +2.1 points, intervals -4.7 to +10.4 and -5.7 to +9.7), so by the declared rule the selected point is retained. The dipole control, however, loses 17–23 points at both alternatives, where the matched centroid passes the magnet about 1 mm farther from the stored orbit, and the NKM-minus-dipole difference falls to -8.0 and -9.4 points. The size of the NKM-specific deficit is therefore a property of the operating point as well as of the kicker; the selected point, chosen without injection errors, favours the uniform kick.

On corrected ring 5501 without injection errors, the NKM and the dipole capture 29.8 and 33.2 %. 95 % of the NKM losses are vertical and occur over many turns, and opening the in-magnet chamber to 15 mm, an infeasible diagnostic, raises capture only to 33.2 and 37.6 %. Uncorrected coupling, for which the lattice has no skew correctors, is consistent with this but is not isolated here.

VIII. PULSED-MAGNET CONSIDERATIONS

The results above apply to an ideal single-pass spatial field. At the selected point ($x_{\mathrm{inj}} = -19.48$ mm, $S = 0.95$), native tracking of the nominal centroid observes positions -2.5, 5.5 and 7.8 mm after turns 1–3 at the NKM centre. The corresponding straight-line full-field kicks are 0.38, 3.32 and 5.30 mrad. Assigning one tenth of the selected point's conditional angular-window width, 0.44 mrad (0.044 mrad), as a residual allowance is a design assumption, the same convention as the thin-map accuracy allowance of section III; it gives first/second-return fractional limits 0.12/0.01.

For a half-sine of total duration τ, injection at its peak ($t = \tau/2$) gives first-return field fraction $\cos(\pi T_0/\tau)$ if $T_0 \leq \tau/2$, and zero otherwise. Later returns replace T_0 by nT_0

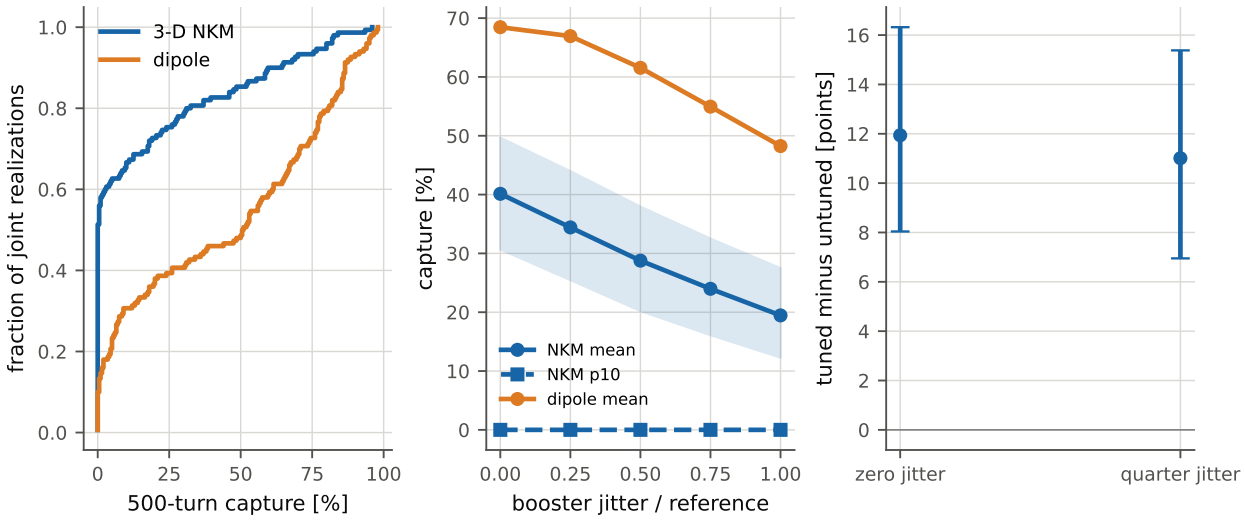


FIG. 10. Final design-beam robustness with the reference chamber. Left: empirical cumulative capture distributions for 150 paired joint-error draws, conditional on ten corrected rings. Centre: NKM mean and tenth percentile and dipole mean versus jitter scale (50 paired draws); shading is a 95 % realization-bootstrap interval of the NKM mean. Right: held-out tuned-minus-untuned mean capture at zero and quarter jitter, with paired 95 % intervals (50 machines, 100 training and 200 held-out particles each). Bootstrap count 10 000, seed 1729. All tracking uses 500 native turns.

in this finite-support rule. The native-centroid allowance gives $\tau \leq 5.76\,\mu$s; zero field on every return requires $\tau \leq 5.33\,\mu$s. A $5.3\,\mu$s peak-phase example meets the latter condition, whereas $8\,\mu$s retains 0.50 of the peak on the first return. Deterministic timing offsets obey $1 - \cos(\pi\Delta t/\tau)$, giving 7.7×10^{-8}–4.9×10^{-6} for 1–5 ns and 5–8 μs. The 2 ns Gaussian rms in the error model is a distribution, not a hard timing bound.

A train with its first bunch at the peak samples less field on later bunches: for the $5.3\,\mu$s example, 100, 200 and 500 ns train durations give approximately 0.18, 0.70 and 4.4 % variation. At an offset kicker the stored-beam centroid response scales with that field; aligned centroid cancellation does not imply zero finite-beam moment change. On-axis residual dipole and gradient allowances are 4×10^{-5} and 7×10^{-5} of the corresponding peak quantities. For the assumed flat chamber, a thin conducting layer with half-gap $a = 3.8$ mm has diffusion time of order $\tau_{\mathrm{eddy}} \simeq \mu_0\sigma a t/2$: a $1\,\mu$m coating with $\sigma = 10^6$ S/m gives 2.4 ns and an $8\,\mu$m titanium coating, as on the ALS ceramic chamber [18], about 46 ns, both short compared with the $5.3\,\mu$s pulse; a 1 mm metallic wall with $\sigma = 5 \times 10^7$ S/m would give 0.12 ms and is excluded. For the ALS chamber a 1.1 % attenuation of the peak field was reported [18]. These

are order-of-magnitude scenarios, not predictions for an unspecified chamber. Coating-induced multipoles require chamber-specific analysis [11]; no time-dependent capture or operational pulse feasibility is claimed.

IX. DISCUSSION AND LIMITATIONS

Finite-length accuracy depends on trajectory displacement relative to the kick-lobe width. Calling a model a kick map does not specify how it treats that displacement. This is consistent with the small finite-conductor effect reported for the ALS kicker [18]: different geometry and launch conditions place that design at a much smaller sagitta, and the result here does not imply that its model was inadequate. The tested sagitta ratio organizes deterministic comparisons over a scaled wire family; it does not establish a universal capture threshold. A sliced vector map retains both the physical vertical field and self-deflection. Local vertical focusing changes sign across the kick extremum. The first-turn vertical losses of the zero-capture realizations come mainly from the uncorrected vertical injection error (section VII C); slower vertical losses on the corrected rings, as on ring 5501, are consistent with uncorrected coupling, without identifying a mechanism.

Ring errors that are corrected in orbit, optics and tunes reduce capture almost equally for the NKM and a uniform kick. The NKM-specific loss arises from injection errors. The paired ablation shows that, under the modelled horizontal-only steering, transfer-line magnet errors leave a static vertical injection error that removes most NKM shots on the first turn, and booster jitter removes largely the same shots. Correcting the vertical injection error, by steering or by transfer-line alignment tight enough to make it negligible, and stabilizing the booster extraction are therefore required in this model. Their hardware implementation is not evaluated here. The two-plane sensitivity is an optimistic benchmark for ideal static centroid removal, without BPM or actuator errors; it does not establish a global upper bound on capture. The size of the deficit also depends on the operating point. The selected point was chosen without injection errors and favours the uniform kick. Selection with injection errors in the objective, and sextupole optimization for injection acceptance, remain open.

Nominal capture, a joint-error mean, the zero-capture fraction and the fraction of realizations above 80 % answer different questions. The ring-cluster interval shows how the mean and lower tail depend on the declared corrected-ring population. The 500-turn window

is a conditional survival contour; its 4000-turn checks quantify later losses. The modelled steering is ideal and corrects two static coordinates (four in the sensitivity); the tuning search uses an ideal capture observable, and its declared range truncates the optimum for a quarter of the machines. Neither establishes a practical feedback tolerance or commissioned efficiency. Small BTS Twiss mismatch alone does not establish successful ring capture.

The modelled transfer line has no vertical correction, so its vertical injection error follows from the declared alignment budget. The ring version, aperture shape, IVU assignments, incoming-line clearance, error budget, the 120-corrector system without skew correctors, and the ten corrected rings limit transfer to hardware. The filament model omits iron, chamber-induced fields, pulse evolution and actual septum leakage. The pulse section supplies consistency estimates, not time-dependent capture. Sextupole optimization may alter the acceptance, as commissioning studies at BESSY II and SIRIUS illustrate [9, 15]; it is outside this study. The joint capture ensembles vary whole-magnet placement and common field scale, not individual wires or loop-current imbalance. Wire tolerances refer to one error at a time, and aligned centroid symmetry does not eliminate finite-beam moment changes or manufacturing errors.

X. CONCLUSIONS

For the supplied KLS design lattice, a single thin kick map does not describe the 0.525 m four-loop NKM at the selected operating point: along its matched trajectory the thin-map error is 3.7 times a tenth of the 0.44 mrad angular window, and for a diagnostic launch family the error is organized empirically by the sagitta-to-kick-width ratio, while a sliced vector map with the consistently signed vertical field agrees with the exact filament field.

With the design booster beam, a feasible flat chamber inside the magnet and an operating point selected on corrected rings, capture is 98.4 % (500 turns) and 97.8 % (4000 turns) on the ideal ring and 78.0 % on average over ten corrected rings. Under the assumed joint errors it falls to 16.1 %, against 44.1 % for a uniform kick. Paired ablation attributes the NKM-specific loss without booster jitter to a static vertical injection error from transfer-line magnet errors that the modelled steering does not remove; with ideal two-plane steering that loss nearly disappears (76.6 %), and booster jitter, in position as well as in angle, becomes the NKM-specific term. The deficit relative to the uniform kick also depends on

the operating point. Under horizontal-only steering and without booster jitter, capture-based tuning recovers about +11.9 points, with 28 % of the machines at the declared range limit.

The two-plane sensitivity is an ideal static-removal benchmark, with no correction hardware or monitor errors modeled. It does not establish a global capture bound.

Transparency requires compensation of the static on-axis residual of a realistically built magnet, and septum leakage must be held near 0.07 G at the stored orbit. The results are conditional on the design lattice, the declared chamber and error budget, and the ten corrected rings.

DATA, SOFTWARE AND CODE AVAILABILITY

The simulation code, configurations and analysis scripts are available in the nkm-injection repository (`https://github.com/cspark7701/nkm-injection`). The figures and numbers are generated from the campaign artifacts listed, with SHA-256 digests, in `figure_provenance.csv` of the manuscript directory `docs/paper/generated`. A frozen referee attachment contains the selected simulation outputs, input files, code snapshot, hash inventory and verified rebuild instructions. It has no public DOI; the repository alone does not contain the ignored simulation outputs.

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